

EMCH 260

SOLID MECHANICS

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PREVIOUS ON CHAPTERS 1&2

Stress

Describe
intensity of
internal
resultant
force

**Relationship
???**

Strain

Describe
material
deformation



Mechanical Properties of Materials

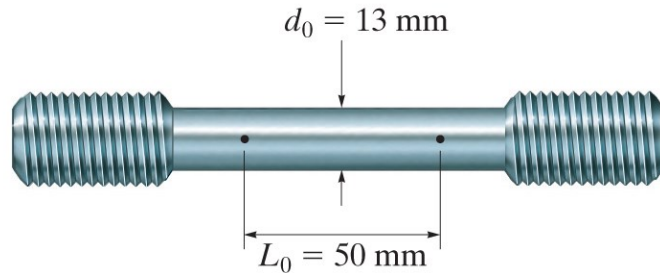
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Chapter 3.1

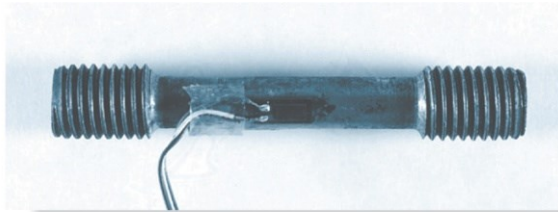
- ✓ Understand how to measure the stress and strain through experiments
- ✓ Correlate the behavior of some engineering materials to the stress-strain diagram
- ✓ Know how to apply modulus of elasticity to determine the deformation of materials

TENSION AND COMPRESSION TEST

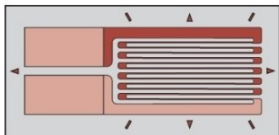
- *To prepare experimental samples*



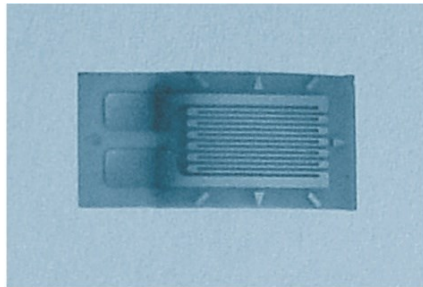
- *To measure material deformation*



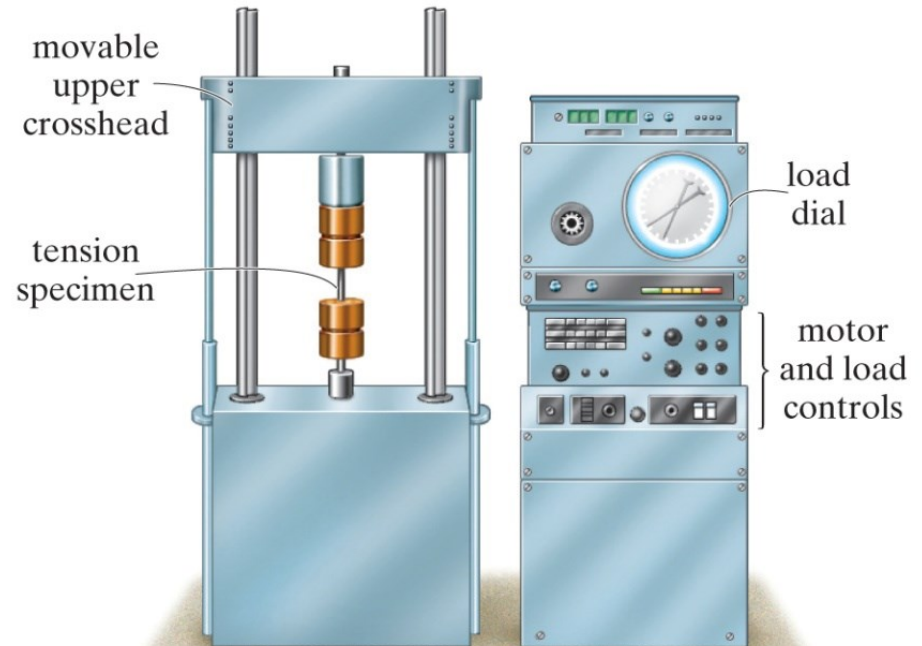
Typical steel specimen with attached strain gauge.



Electrical-resistance strain gauge



- *To apply external loadings (tension or compression)*



ENGINEERING STRESS AND STRAIN

- Nominal or engineering stress

$$\sigma = \frac{N}{A_0} \quad A_0: \text{original cross-sectional area}$$

- Nominal or engineering strain

$$\varepsilon = \frac{L - L_0}{L_0} = \frac{\delta}{L_0} \quad L_0: \text{original gage length}$$

-
- True stress

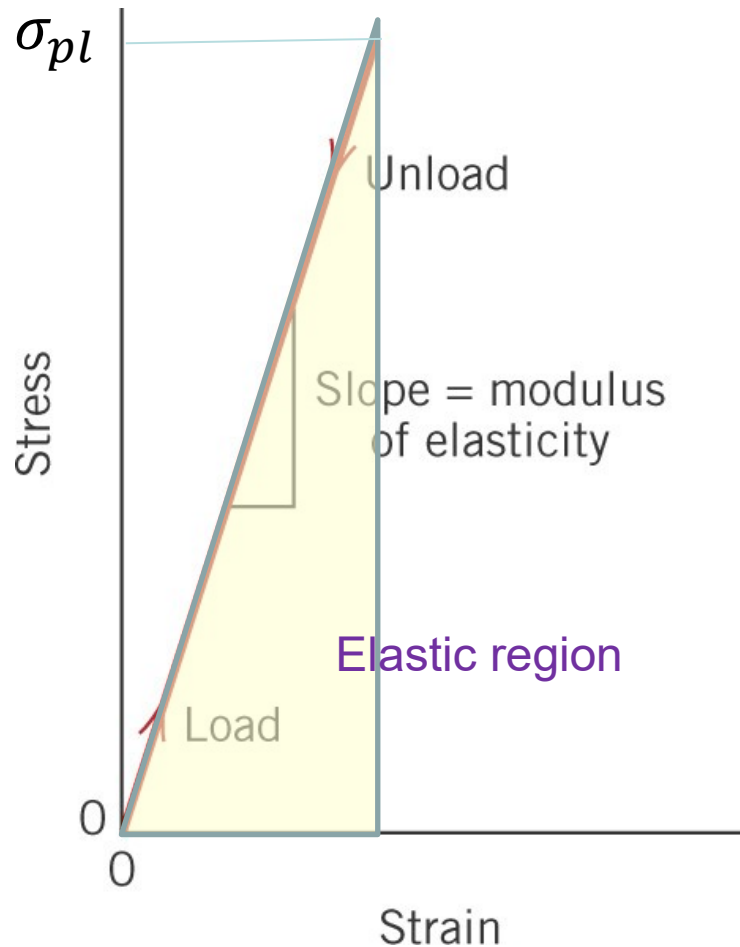
$$\sigma = \frac{N}{A} \quad A: \text{actual or instant cross-sectional area}$$

- True strain

$$\varepsilon = \frac{\delta}{L} \quad L: \text{actual or instant gage length}$$

ELASTICITY & YOUNG'S MODULUS

(proportional limit)



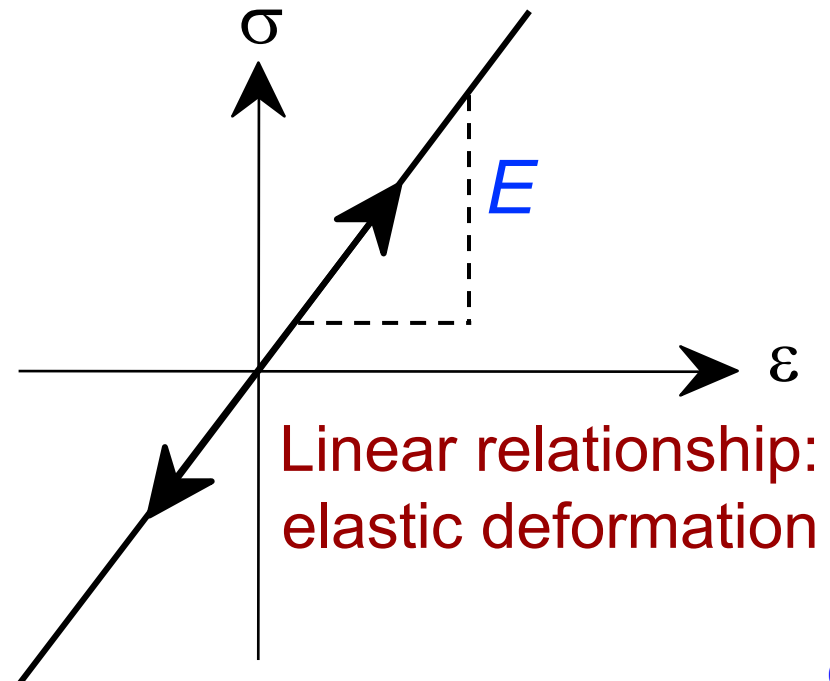
- Modulus of elasticity or Young's modulus (E)

also indicates the **stiffness** of a material

- Hooke's Law:

$$\sigma = E \epsilon$$

Usually in unit of GPa (10^9 Pa)



ELASTIC DEFORMATION UNDER LOADING

- Combine Young's modulus (E) and definition of stress and strain

$$\sigma = E \cdot \varepsilon$$

$$\sigma = \frac{N}{A}$$

$$\varepsilon = \frac{\delta}{L_0}$$



$$\delta = \frac{NL_0}{EA}$$

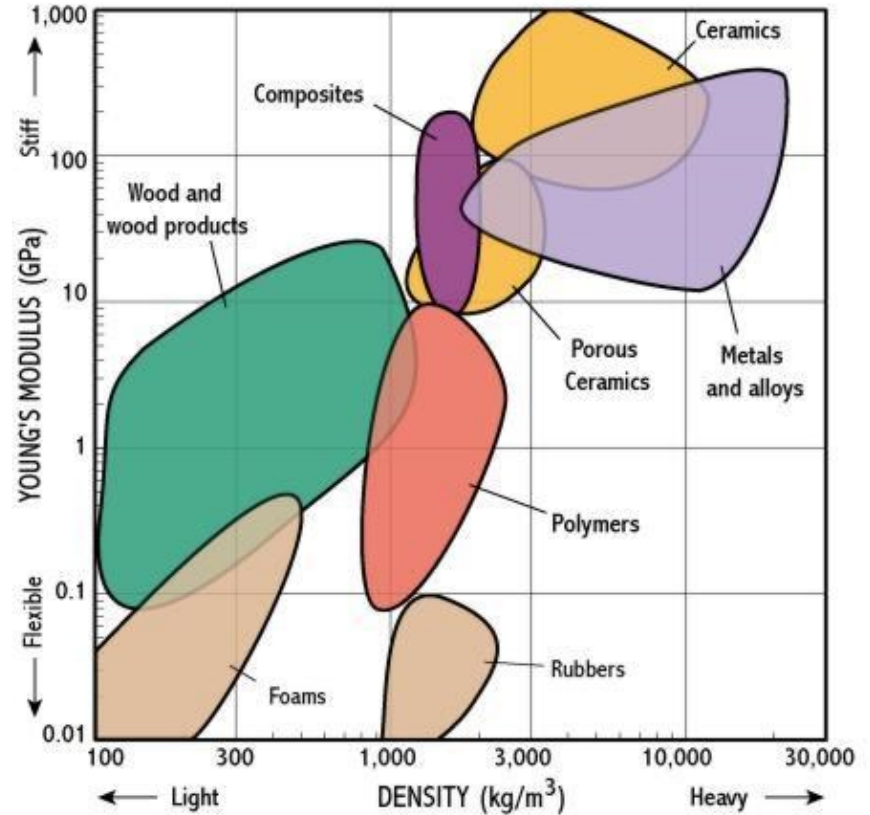
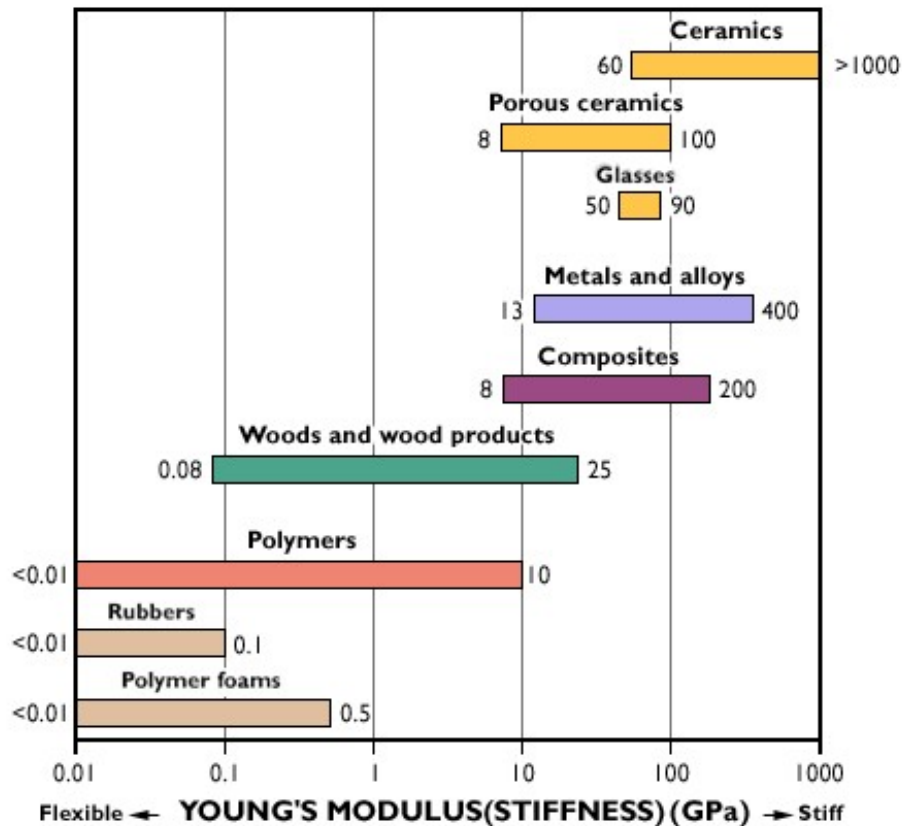
**Elastic deformation under
tensile or compressive
loading / force**

L_0 : initial length of the material

A : cross-sectional area of the material

N : external tensile or compressive loading / force (perpendicular to the cross-section of the material)

YOUNG'S MODULUS OF MATERIALS

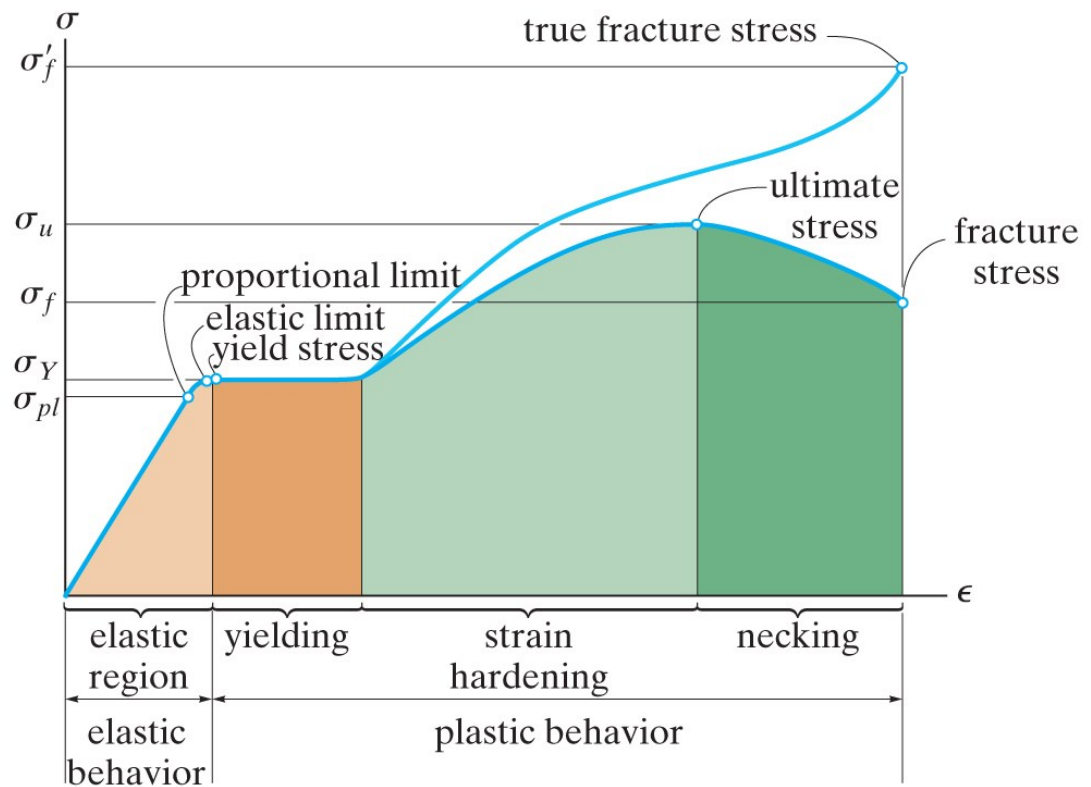


<http://www-materials.eng.cam.ac.uk/>

CONVENTIONAL STRESS-STRAIN DIAGRAM

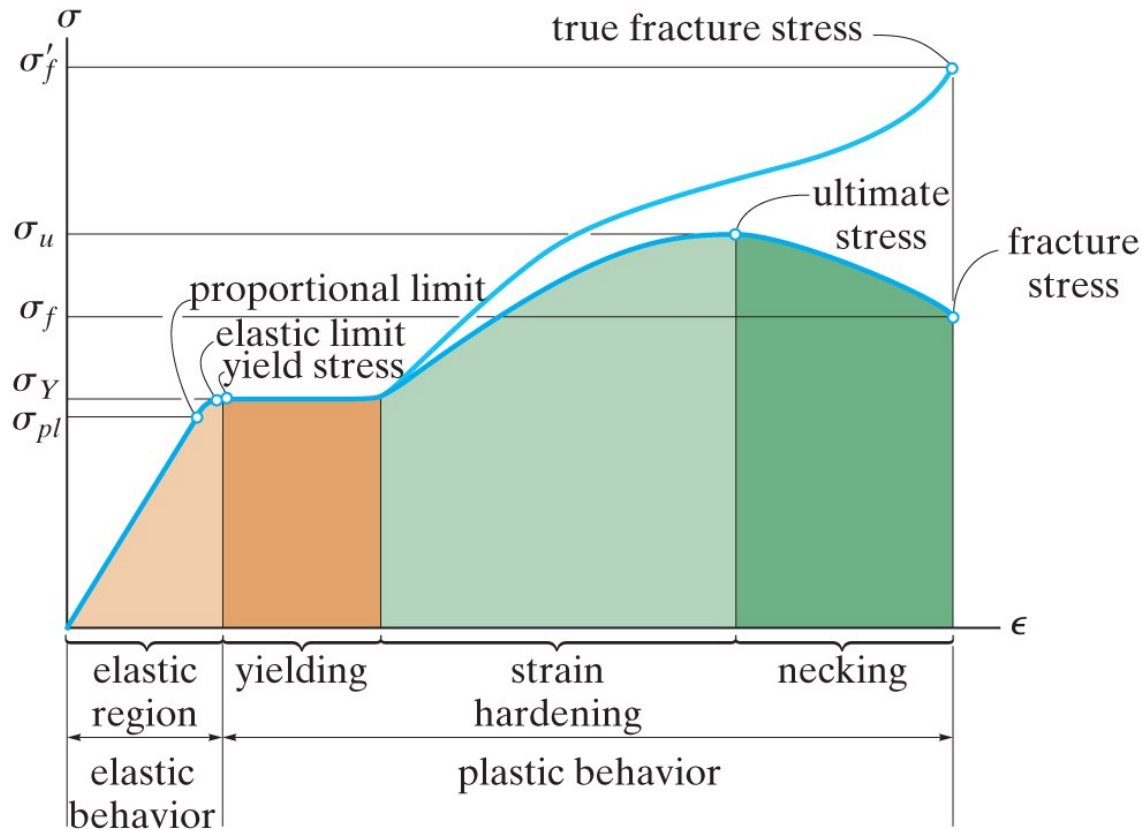
- Note the critical status for strength specification

- Proportional limit
- Elastic limit
- Yield stress (σ_Y)
- Ultimate stress
- Fracture stress

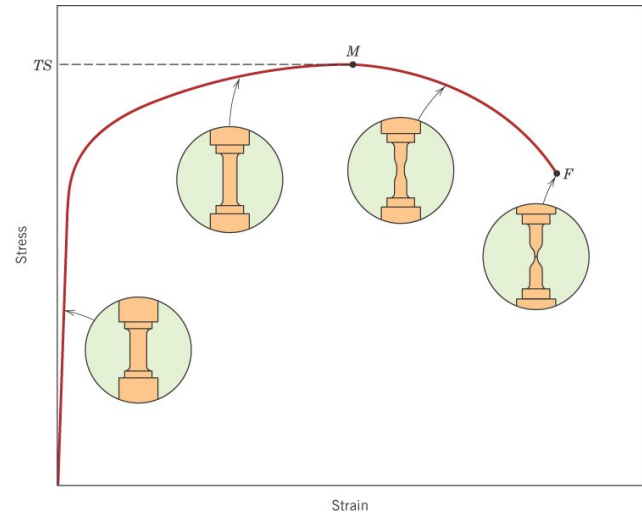


Conventional and true stress-strain diagrams for ductile material (steel) (not to scale)

ULTIMATE STRESS & FRACTURE STRESS



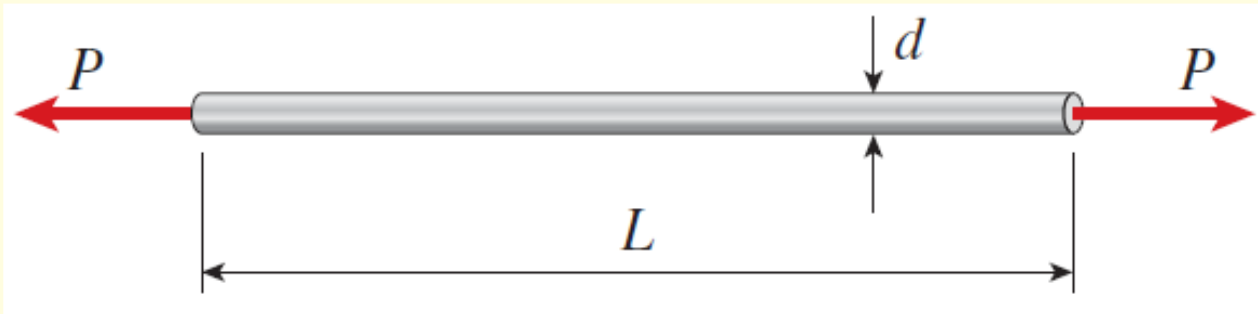
Conventional and true stress-strain diagrams for ductile material (steel) (not to scale)



This steel specimen clearly shows the necking that occurred just before the specimen failed. This resulted in the formation of a “cup-cone” shape at the fracture location, which is characteristic of ductile materials.

IN-CLASS EXAMPLE

A prismatic bar with a circular cross section is loaded by tensile forces $P = 65$ kN. The bar has initial length $L_0 = 1.75$ m and diameter $d = 32$ mm. It is made of aluminum alloy with modulus of elasticity $E = 75$ GPa. Find the increase in length of the bar.



Logic:

Length or geometry $\leftrightarrow$ strain $\leftrightarrow$ stress $\leftrightarrow$ external force

Stress-strain behavior/relationship $\leftrightarrow$ Modulus of elasticity (Young's modulus)

IN-CLASS EXAMPLE (cont'd)

Solutions

- Internal stress of the bar:

$$\sigma = \frac{P}{\frac{1}{4}\pi d^2} = \frac{65000}{\frac{1}{4} \times 3.1415926 \times 0.032^2} = 8.082 \times 10^7 \text{ Pa}$$

- Applying Hooke's law $\sigma = E \varepsilon$, the strain of the bar is:

$$\varepsilon = \frac{\sigma}{E} = \frac{8.082 \times 10^7}{75 \times 10^9} = 1.0776 \times 10^{-3} \text{ mm/mm}$$

- Applying the definition of tensile strain $\varepsilon = \frac{\delta}{L_0}$, the increase in length of the bar is:

$$\delta = \varepsilon \cdot L_0 = 1.0776 \times 10^{-3} \times 1750 = 1.886 \text{ mm (Ans.)}$$

RELEVANT PRACTICE

- Assignment 1, problem #5 (go to Blackboard under the “Assignments” folder, **Assignment 1 due: 8:30 AM, Monday, Sept. 13th 2021**)
- Optional Extra Exercise 1, problem #3-4 (go to Blackboard under the “Assignments” folder, **Assignment 1 due: 8:30 AM, Monday, Sept. 13th 2021**)
- *Submit your solutions directly onto Blackboard*

OUT-OF-CLASS READING

- **Textbook, pp. 85 – 90**